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Construction defects, envelope degradation and lifecycle carbon implications for building energy performance: evidence from Lagos, Nigeria

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ABSTRACT

Little attention has been given to the quality of construction as a potential influence on the energy and carbon performance of buildings in many developing economies. In addition to causing post-construction defects that compromise building envelopes, poor construction quality reduces the long-term operational efficiency of building materials. In this paper, we reconceptualize construction defects as indirect drivers of energy and carbon use by integrating defect severity, repair frequency, and lifecycle considerations into a Quality-Sustainability Integration Index (QSII). We employed a mixed-methods approach comprising a survey of 83 construction professionals in Nigeria and a longitudinal review of maintenance records for 14 residential, commercial, and institutional buildings in Lagos, Nigeria. BIM-based Lifecycle Assessment was used to quantify the embodied carbon associated with defect-induced repair cycles, and a conceptual Digital Twin prototype was developed to visualize defect propagation pathways and intervention strategies. Results indicate that envelope- and waterproofing-related defects generate disproportionately high lifecycle carbon emissions and indirect operational energy penalties due to thermal degradation in hot-humid climates. Findings are calibrated for Lagos's hot-humid climate and institutional context; application to other climatic or regulatory settings requires local empirical validation and parameter recalibration. By positioning construction quality as a latent lever for energy efficiency rather than solely a management issue, this research advances building performance discourse beyond the design stage and emphasizes defect prevention for long-term sustainability.

1. Introduction

The United Nations Environment Programme [1] states that 37% of all global carbon emissions come from buildings' construction and ongoing maintenance. The significant amount of research on building energy has mainly focused on design phase optimization, and there is relatively little empirical evidence about how buildings perform after completion. Buildings rarely operate under ideal conditions because the envelope of a building, the management of moisture, continuity of insulation, and air-tightness of the building are often negatively impacted due to poor workmanship, lack of maintenance, and recurring construction defects. All of these problems will reduce the thermal performance and drastically alter the total carbon footprint throughout the life cycle of the building.

In the case of Lagos Nigeria, a rapidly developing megacity situated in a hot-humid tropical environment, building envelope failures occur because of inadequate waterproofing, insufficient concrete cover, poor roof detailing, and discontinuities in the thermal barrier, leading to increased moisture intrusion and conductive heat transfer, which increases demand for cooling, accelerating premature material replacement. The result is substantial added embodied carbon from repair processes as well as indirect operational energy costs. However, the overwhelming majority of current building energy codes measure operational performance only at their initial occupancy and do not factor in ongoing deterioration of the building envelope or lifecycle carbon emissions. This creates a large gap between projected theoretical energy use and actual in-use performance, particularly for new economies, which do not have sufficient monitoring capabilities.

Quality management systems in construction have typically classified defects as either technical failures or maintenance costs [2,3] and assessed their severities based on structural integrity and the costs of rectification without considering the energy or carbon costs involved. Okuntade [4] has developed a systematic taxonomy of 29 types of defects, ranking them with respect to the Nigerian construction industry.

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Yet, like most other defect management studies, this methodology does not quantify either carbon or energy involved in respect of the repeated repair cycles and degradation of the envelope. In the context of limited data for the evaluation of long-term performance, this inconsistency is especially significant in terms of identifying the energy or carbon impacts of any defect (Table 1).

The purpose of this research is to provide an integrative framework for evaluating both the quality aspect of a construction defect and its sustainability consequences via employment of a new composite metric called Quality-Sustainability Integration Index (QSII). QSII expands upon the work of Okuntade [4], who used the severity of defects to evaluate defects according to their lifecycle carbon footprint. QSII provides a composite metric of sustainability by taking into account additional variables such as repair frequency and carbon emissions from the complete building lifecycle. The QSII was developed through a combination of primary data collected from 83 construction professionals during a survey and an analysis of building maintenance records for 14 buildings over five years in Lagos, Nigeria. The buildings also underwent carbon modelling using building information modelling (BIM). In addition to developing the QSII, a conceptual digital twin prototype was developed to visually represent ways that defects propagate (i.e., how they develop over time following original detection) and evaluate interventions for programmatic processes. This study's primary focus on Lagos provides much-needed context for developing context-specific metrics of defect-related carbon emissions for construction systems located in a hot-humid climate and consists of unique features such as institutions, material-specific supply chains, and construction methods. There are three specific goals of this study. First, to expand the theory of defect severity to include lifecycle carbon and envelope performance; second, to determine the lifecycle carbon multiplier for common defect types in Lagos; and third, to demonstrate use of a quality-sustainability assessment method that works well in emerging economies with limited access to complete data. The remainder of this article will be organized according to sections/parts. Sections 2 provides a review of applicable literature and theoretical models; Section 3 explains the methods used to collect data regarding defect severity and lifecycle carbon emissions; Section 4 discusses the findings and modelling; Section 5 analyzes the contributions and implications of this research towards theory and practitioner; and finally, Section 6 provides recommendations for further research.

2. Literature review & theoretical framework

2.1. Construction defects: from technical failures to performance degradation

Traditionally construction defects have been defined in very narrow technical terms. For example, Ramson [5] states that any deviation from intended design or specified quality or workmanship standards can constitute a defect. The primary way in which to evaluate this defect type is according to the cost of remediating the defect. Seeley [6] also provides an extension of Ramson's definition of construction quality by correlating construction quality to both future maintenance requirements and service life performance. In his landmark paper on construction defects in the Nigerian context, Okuntade [4] provided a systematic listing of 29 different types of construction defects based on a consensus of professional opinion and established a hierarchy of severity of each type with the type having insufficient cover over reinforcing of concrete rated as the most severe type of defect due to its effects on the durability of a structure. However, like most construction defect management research, the research conducted by Okuntade only examines and defines defects from technical and economic perspectives, without considering the environmental externalities associated with these defects.

In more recent research, a systemic perspective began to develop. Atkinson [7] found that nearly 82% of construction defects are caused by management or organisational factors (as opposed to simply technical issues), indicating that governance-level intervention is key. In the Nigerian institutional setting, Fagbenle and Oluwunmi [8] have identified inadequacies in quality assurance systems as a contributing factor in building collapse rates nearly 24 times higher than those of OECD countries. Although they have studied the governance aspects of the problem, none of the studies identified a quantitative relationship between defect recurrence and CO_2 emission and/or envelope energy performance degradation over the operational life of a building. The lack of a connection between defect severity and long-term carbon and energy consequences is the most significant gap present in hot-humid climates, where envelope failures result in moisture intrusion, reduced thermal resistance, and hastened material deterioration, which both create embodied carbon (due to repair cycles) and generate operational energy penalties (due to poor envelope performance).

2.2. Sustainability assessment and lifecycle carbon in building performance

Building sustainability assessment frameworks have changed from

Table 1
Summary of key literature on construction defects, lifecycle assessment, and digital integration with identified research gaps.

Study/source	Focus area	Key contribution	Research gap addressed
Okuntade [4]	Construction Defects	Ranked 29 defect types by severity in Nigerian context	No lifecycle carbon or energy metrics included
Atkinson [7]	Construction Quality	82% of errors stem from managerial failures	No quantification of carbon/energy implications
Fagbenle and Oluwunmi [8]	Building Failure – Nigeria	Institutional failures drive collapse rates 24 × OECD	No connection to thermal performance or LCA
Liu et al. [12]	Digital Twin/BIM	Performance gap between theoretical and actual operation	Assumes idealised build quality; excludes defect propagation
Buyle et al. [22]	Lifecycle Assessment	Comparative LCA methodologies in construction	LCA models omit defect-induced material turnover
Pomponi & Moncaster [20]	Circular Economy/LCA	Carbon multipliers and material circularity	Limited application in data-constrained contexts
Lu et al. [13,15]	BIM & Digital Twin	BIM for FM; Digital Twin for sustainable environment	Assumes complete data; resource-constrained contexts neglected
Yang et al. [16]	Passive Retrofit/Hot-Humid	Energy performance of courtyard dwellings across climates	No defect propagation pathway analysis
Yang et al. [17]	Multi-Objective Optimization	Green renovation of Tibetan courtyard dwellings via NSGA-II	Different climate but methodology contextualises QSII approach
Zhang et al. [10]	Quality–Sustainability	Systematic review of construction quality and sustainability	Calls for integrated frameworks in developing economies
This Study	Defects + LCA + Energy	QSII integrating severity, carbon impact, repair frequency	Construction quality as latent energy/carbon lever in Lagos

concentrating primarily on operational energy efficiency to include more holistic lifecycle of the building and consider all aspects related to embodied carbon, material circularity and climate resiliency [9,10]. The ISO14044 [11] is the basis for establishing a normative framework regarding Lifecycle Assessment (LCA). One of the continuing flaws with currently available LCA applications involves the belief that building performances are consistent; most models do not consider the potential for performance degradation due to defects or the recurring carbon emissions created by the need for continual repair, thus causing a gap between predicted design stage performance and actual performance over time [12].

Data integration for all lifecycle phases has been furthered through advancements in digital tools. Building Information Modelling (BIM) facilitates the linking together of performance data across all of the phases of the building's lifecycle [13], whereas the concept of a Digital Twin provides a dynamic representation of a building's performance over time [14]. The majority of digital performance modelling approaches make assumptions regarding ideal builds, and have not taken into account defect propagation pathways or maintenance-related material turnover, and as noted by Lu et al. [15], digital twin applications are still methodologically undeveloped in resource-constrained environments.

There are great similarities between the recent developments made in building performance analysis and the methodology described here. As an example, a study by Yang et al. [16] assesses the performance of different types of retrofit of courtyard dwellings in China, with specific reference to their location in hot-humid climates. The results indicate that retrofits of courtyard dwellings – that have similar features to the ones assessed in this study – may be subject to substantial energy penalties in hot-humid locations due to increased cooling capabilities; these results provide important context for the focus of this project on determining retrofitted courtyard dwellings in hot-humid areas and how that will affect their thermal performance. In another study by Yang et al. [17], a framework for multi-objective optimization of the green renovation of Tibetan courtyard dwellings has been developed using NSGA-II; by employing Sobol' sensitivity analysis and optimization with back propagation neural networks (BPNN) to reduce energy use, thermal discomfort hours and initial cost simultaneously, the researchers developed a composite indicator methodology that provides useful comparison for the QSII's multidimensional assessment framework. Further, both studies emphasize the necessity of calibrating building performance assessments to the climate where the building is located, thereby supporting the justification for creating context-specific indices, like the QSII, rather than applying globally generic frameworks.

2.3. Theoretical positioning: governance, circularity and carbon multiplier mechanisms

This research combines the Transaction Cost Economics (TCE) and the Circular Economy theory to form the theoretical basis of the quality-sustainability relationship. Williamson [18] has shown within TCE that defects in initial quality cause additional transaction frequency due to maintenance actions being repeated and information asymmetries existing between the parties involved in the construction process. In construction, repeated deficiencies produce hidden governance costs which materialise as repeated repair contracts and the use of supervisory intervention representing greater than just financial costs through both material waste and total embodied carbon created during the defect lifecycle. Circular Economy principles [19,20] provide additional context to support characterized repair cycles induced by defects as interruptions to circularity of materials as each repair cycle creates embodied carbon and reduces the effective service life of the component. Therefore, investments in quality can be reconceptualized to be interventions that support the achievement of circularity by extending service life, decreasing material turnover and preventing total embodied carbon accumulation.

This research introduces the carbon multiplier concept as a formal way to express the non-linear link between a defect and its overall carbon footprint. Instead of being linear models where all repair carbon comes only from the number of materials required to complete the repair (e.g., waterproofing), carbon multiplier models have a cascading impact due to other aspects of the failure mechanism. For example, if a waterproofing defect fails, it can cause secondary damages to the ceiling system, structural elements and electrical systems, which in turn will all require additional materials (and, therefore, carbon) to remediate. As a result, the carbon multiplier model provides a more comprehensive basis for prioritising quality investment than merely looking at structural severity criteria alone, as it provides a more complete view of the defect's downstream carbon liability.

3. Methodology

3.1. Research design and philosophical approach

The study adopts a pragmatic epistemological approach, using a mixed methods research design. It collects quantitative survey data, conduct a case study analysis, and create a conceptual model. The design philosophy acknowledges that construction knowledge exists within both social and cultural contexts, and therefore seeks to use established local methodologies that are compatible with Nigerian practice rather than applying an externally developed framework. This is termed as 'contextual theoretical extension' and builds incrementally on Okuntade's [4] validated framework of defect severity, adding dimensions of sustainability but retaining empirical validity for the Nigerian context.

A three-step process is used in this research. Phase One uses a structured survey of professionals to obtain data on the types of defects and their severity (this is a repeat of Okuntade's [4] methodology). Phase Two uses a longitudinal study to determine how defects propagate from one item to another and what materials were used to repair them. Phase Three integrates the findings from both previous phases into a composite Quality Assurance for Information Technology (QAIT) metric and a conceptual Digital Twin prototype. The results of each phase are used to impact the other phases, i.e., the calibration of the survey informs the parameters for the Life Cycle Assessment (LCA) model; how defects propagate informs the architecture of the Digital Twin's defect scenario.

3.2. Data collection procedures

3.2.1. Quantitative survey

A structured questionnaire survey was conducted with construction professionals working in Lagos to replicate Okuntade's [4] defect severity framework and extend it to include dimensions of sustainability. The survey serves as a confirmation tool for the parameters of professional judgement in terms of determining defect severity, carbon footprint, and reoccurrence of repairs for the analytical framework, rather than a representative sample of the population. Participants were recruited from professional networks containing builders, architects and civil engineers with direct experience in the delivery of construction projects, supervising construction projects or maintaining construction projects. A minimum of three years' experience as a professional was required to participate in the study.

Data were collected in the form of 83 valid responses received over a period of January to April 2023, giving rise to a total response rate of 72%. The total sample consisted of 36 builders (43.4%); 22 architects (26.5%); and 25 civil engineers (30.1%). The survey included all 29 categories of defects identified by Okuntade in 2014 for methodological consistency; therefore, each respondent evaluated traditional defect severity on a four-point scale, carbon emission impact judgment on a five-point ordinal scale, and annual repair frequency estimates. Survey results provided calibration for the QSII's Severity Index (SI) component, Carbon Emission Index (CEI) component, and not distributions

from which inferences may be made about the entire population based upon statistical evidence.

3.2.2. Case study analysis

Collection of maintenance records that could count towards the overall performance of the given buildings in Lagos was done at 14 buildings represented by a total of eight residential, four commercial and two institutional buildings, with formal consent obtained from each building's facility manager. All locations have been constructed within the last five years (between 2015 and 2018) and have a minimum of three years of records. In instances where there is a larger set of records available (5 years of records – 2018–2023 was preferred), they were used. A representative sample of buildings were selected for variance in type of building (residential, commercial or institutional), quality of construction in order to achieve a broad range of analysis at all levels of typology and performance level, as described previously. Due to the unavailability of Environmental Product Declarations (EPDs) specific to Nigeria, the amount of embodied carbon was determined using the University of Nottingham's (ICE v3.0) emission factors as baseline relative to the carbon emissions for each of the buildings examined. The results of the sensitivity analysis that altered the emission coefficients (±20%) confirmed that the aggregate rankings of all the defect categories were stable throughout. To quantify embodied carbon, standard LCA reference databases were applied. Given the absence of Nigeria-specific EPDs, ICE v3.0 emission factors were used as relative multipliers; detailed limitations and uncertainty mitigation strategies for these proxy factors are provided in Section 3.2.4.

3.2.3. Data completeness and estimation procedures

Data completeness summary: A total of 847 repair entries were reviewed. Approximately 18% of repair entries (153) required partial material quantity estimation. Approximately 7% of repair entries (59) required complete reconstruction of both quantity and cost data. Flagged entries (reconstructed) are provided. As required for full transparency of methodology, these results are provided for ease of identifying the repairs needing these methods.

The maintenance documentation quality was inconsistent throughout the 14 buildings studied. Frequency of repairs for each building was consistently reported; however, there were some entries without documented quantities of materials used. For all entries missing quantities, an estimated quantity was calculated using the median documented quantity of materials used for repairs that are representative of the project type (i.e., residential, commercial, or institutional). Any entries with missing quantities or costs were reconstructed using descriptions of defects, records of work performed, and material specifications obtained from similar repair projects in the same data set.

Estimations of parameters that were reconstructed were used to generate regular distributions rather than single points to avoid inappropriate statement of the precision of an estimation. All estimations of uncertainty are carried forward in a way that makes it clear to all users of the analytical model that uncertainty related to documentation is carried transparently through the analytical model rather than being absorbed covertly into single point estimates.

3.2.4. Expert consensus validation workshops

In Lagos, 17 seasoned professionals (architects, engineers and contractors) participated in three structured validation workshops to reach consensus on established backgrounds related to building performance and assessing defects to determine their weight in the QSII. The single round of consensus validation was conducted with the use of a structured process different than the Delphi survey method consisting of one round. The workshops were used as an opportunity to validate survey findings and calibrate the QSII weighting coefficients. The participants each assigned relative weights to each of the three QSII criteria based on a maximum total weight of 1.0; the final calibrated weights were based on the median of all individual allocations (weights assigned by all participants). The weights validated through consensus were: $W_1 = 0.50$ (severity), $W_2 = 0.30$ (carbon output), $W_3 = 0.20$ (repair frequency).

3.2.5. Limitations of proxy emission factors.

The Inventory of Carbon and Energy (ICE) v3.0 database has been used for embodied carbon calculations because there are no Nigeria-based Environmental Product Declarations (EPDs). ICE v3.0 does have peer-reviewed and internationally referenced emission factors; however, this particular database also has some important limitations that should to be stated explicitly:

(1) Regional Differences in Materials: Nigerian cement blends are more often produced with higher levels of local supplementary materials (e.g., calcined clay or limestone fill) than European formulations that form the foundation for the ICE factors, which could lead to a variance of +/- 15 to 25 percent in the carbon footprint of the produced cement [21].
(2) Supply Chain and Transportation Differences: Local energy supplies, alternative sources of materials to find, and methods of moving materials around Lagos will affect lifecycle emissions when compared to standard industrialized production assumptions.
(3) Imported Coefficient Uncertainty: Applying coefficients derived from Global North contexts to emerging economies risks systematic bias when production systems, grid intensities, and material standards differ substantially [20].

The limitations of this study were alleviated using: (a) ICE factors used only as relative multipliers for comparative defect rankings [22] instead of absolute carbon accounting; (b) partial to full sustainability variation of ± 20% for emission coefficients to demonstrate stability of relation between defect categories; and (c) all carbon values indicated but not definitive. We hope that the creation of Nigerian EPDs within specific contexts will be developed as tools to support timely lifecycle assessment that may occur through research.

3.3. Analytical framework

3.3.1. Extended severity index and weight calibration

The QSII expands Okuntade's [4] Severity Index methodology by incorporating carbon impact and repair frequency dimensions into a composite sustainability metric:

$$QSII = (SI \times W_1) + (CEI \times W_2) + (RFI \times W_3) \quad (1)$$

where:

SI = Traditional Severity Index [4], calibrated from professional survey responses on a 0–100 scale.

CEI = Carbon Emission Index (0–100 scale), derived from expert assessment of carbon impact severity.

RFI = Repair Frequency Index, normalised to a 0–100 scale using min–max normalisation across observed defect frequencies, ensuring dimensional consistency with SI and CEI.

W_1, W_2, W_3 = weighting coefficients ($\Sigma W = 1$), validated through structured consensus workshops: $W_1 = 0.50$, $W_2 = 0.30$, $W_3 = 0.20$.

Robustness testing varied each weighting coefficient by ±20% while maintaining $\Sigma W = 1$, confirming that the top five defect category rankings remained stable across all tested weight combinations.

3.3.1.1. Operational energy penalty estimation approach. Operational energy penalties associated with envelope degradation were not directly simulated using dynamic building energy modelling software. Instead, thermal performance degradation and corresponding cooling load increases were conceptually inferred using established building physics principles and Lagos-specific climate data. Specifically, we applied published U-value degradation coefficients from ASHRAE Fundamentals (2021) to the defect categories identified in the survey and case studies,

and coupled these with Lagos cooling degree-day metrics [23] to estimate incremental operational energy demand. This proxy approach follows established practice for data-constrained contexts where full energy simulation inputs (e.g., detailed HVAC specifications, occupancy schedules, and airtightness test results) are unavailable [16]. All operational energy estimates are explicitly framed as indicative directional impacts rather than precise quantifications, and are integrated into the QSII and scenario analyses as conceptual multipliers to contextualize long-term sustainability liabilities.

3.3.2. Conceptual digital twin development

A conceptual Digital Twin prototype was developed as a descriptive visualization and scenario exploration tool to parameterize defect propagation pathways and compare lifecycle carbon trajectories under different construction quality scenarios. The prototype integrates BIM geometry with lifecycle carbon estimation and Monte Carlo uncertainty analysis (5,000 iterations) to generate probabilistic carbon projections reported as 95% confidence intervals. Three high-impact defect categories identified in the survey and case study phases (insufficient concrete cover, poor waterproofing, and inadequate thermal envelope detailing) were parameterized to simulate degradation processes under Lagos climate conditions. Consistent with Grieves and Vickers [14], this conceptual Digital Twin prototype serves strictly as a conceptual support tool for stakeholder communication and intervention scenario comparison; it is not validated for real-time operational control or predictive energy simulation due to current data infrastructure limitations in the Nigerian context. All uncertainty arising from reconstructed maintenance records is explicitly propagated through the Monte Carlo framework to maintain methodological transparency.. Fig. 1 illustrates the integrated methodological framework, showing the flow from data collection phases through the QSII analytical structure to Digital Twin scenario outputs.

4. Findings

4.1. QSII rankings and carbon multiplier analysis

Okuntade [4] classifies defect severity traditionally; however, these rankings do not align with the QSII results when sustainability criteria are introduced. For instance, while inadequate concrete cover (SI = 83.7) was classified as the most severe defect based on structural durability, the highest composite QSII score was assigned to inadequate waterproofing and drainage (QSII = 86.2). This shift reflects the substantial carbon multiplier (3.4×) attributed to waterproofing failures and their elevated repair frequency. Specifically, initial moisture intrusion initiates a cascade of secondary failures across ceiling, structural, and mechanical systems, each requiring additional repairs that generate substantial embodied carbon. The full QSII rankings for the top 10 defect categories are presented in Table 2.

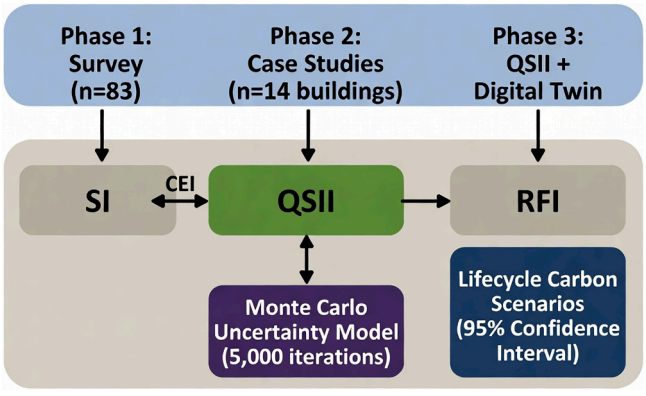


Fig. 1. Integrated Methodological Framework. QSII computation to Digital Twin scenario analysis.

It is important to clarify that the operational energy penalties referenced throughout this study were conceptually inferred from established building physics principles and Lagos climate data, rather than directly simulated via dynamic energy modelling software. Using published U-value degradation coefficients and cooling degree-day metrics, we estimate that envelope-related defects likely increase cooling demand by 12–18% in hot-humid conditions, consistent with climate-adaptive proxy methods [16]. These estimates are presented as indicative directional impacts to contextualize the total sustainability liability of carbon-amplifying defects.

Statistical validation confirmed strong inter-professional consensus in severity assessments: Kruskal-Wallis analysis revealed no statistically significant differences in defect severity ratings across professional groups (builders, architects, engineers): $H(2) = 1.24$, $p = 0.54$, consistent with Okuntade [4]. Spearman's rank-order correlation indicated a moderate, statistically significant positive relationship between traditional SI scores and composite QSII rankings ($\rho = 0.68$, $p < 0.01$), confirming substantial overlap while also highlighting meaningful divergence where sustainability criteria reorder priorities. For example, poor thermal envelope detailing ranks 4th in the QSII with a carbon multiplier of 3.1×, representing a high-recurrence, carbon-amplifying defect that remains obscured in traditional structural severity rankings. All statistical analyses were conducted in R v4.2.1; effect sizes and 95% confidence intervals are reported alongside exact p-values to support interpretability and reproducibility (Cumming, 2014).

Taken together, these results empirically demonstrate that sustainability-informed defect prioritization cannot be reliably inferred from structural severity rankings alone, particularly for moisture-management and thermal envelope defects in hot-humid climates such as Lagos.

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Table 2
QSII rankings — top 10 defect categories with Carbon Emission Index (CEI), normalised Repair Frequency Index (RFI), composite QSII score, and carbon multiplier values. SI = traditional Severity Index; CEI = Carbon Emission Index; RFI = Repair Frequency Index (normalised 0–100).

Rank	Defect category	SI	CEI	RFI (Norm.)	QSII	Carbon multiplier
1	Inadequate waterproofing & drainage	76.4	84.7	64.0	86.2	3.4×
2	Insufficient concrete cover	83.7	71.2	48.5	83.7	2.1×
3	Non-compliance with specifications	74.1	77.8	55.2	79.4	2.6×
4	Poor thermal envelope detailing	68.9	80.3	58.0	78.1	3.1×
5	Premature formwork removal	72.6	68.4	52.3	75.3	1.8×
6	Inadequate roof detailing	70.3	74.1	44.6	73.9	2.4×
7	Poor surface finishing	65.2	58.7	60.1	69.8	1.6×
8	Defective electrical installation	61.8	55.3	71.4	68.4	1.9×
9	Inadequate foundation drainage	69.4	62.1	38.2	67.3	2.2×
10	Substandard material substitution	63.1	66.8	41.5	66.1	1.7×

poor thermal envelope detailing ranks 4th in the QSII with a carbon multiplier of 3.1×, representing a high-recurrence, carbon-amplifying defect that remains obscured in traditional structural severity rankings. Taken together, these results empirically demonstrate that sustainability-informed defect prioritization cannot be reliably inferred from structural severity rankings alone.

4.2. Case study analysis: defect propagation pathways

A longitudinal assessment of the 14 buildings in Lagos showed the same defect propagation patterns, giving clarity to how sustainability liability flows over time. Table 3 illustrate examples of both the archetypal 'pathway' for waterproofing failure, and a pathway for a latent deficiency in concrete cover. Both examples show clear differences in the timing of manifestation and total carbon impact. To quantify embodied carbon, standard LCA reference databases were applied. Given the absence of Nigeria-specific EPDs, ICE v3.0 emission factors were used as relative multipliers; detailed limitations and uncertainty mitigation strategies for these proxy factors are provided in Section 3.2.4.

The pathway for treated waterproofing generated 82.3 tCO_2e for every 1,000 m^2 over five years through the cascade of secondary damage due to the absence of waterproofing treatment at the treated waterproofing location and the concrete cover deficiency generated 67.8 tCO_2e for every 1,000 m^2, but resulted in a much longer delayed expression. The longer delayed manifestation creates a “false economy” disparity between the appearance of defective characteristics in the initial occupancy periods and the resulting sustainable liabilities as the defective characteristics reach their way to full manifestation. The entire sample from the method-based case study determined that buildings with documented defective waterproofing produced up to a minimum of 28% overall life cycle carbon emissions than otherwise similar buildings without documented defective waterproofing (i.e. based upon the quality control method).

Fig. 2 shows the defect propagation and carbon accumulation pathway schematically, illustrating how initial envelope failure branches into multiple secondary damage chains over the five-year observation period.

4.3. Scenerio analysis using the conceptual digital twin prototype: lifecycle carbon scenarios

The conceptual Digital Twin prototype was used to model comparative lifecycle carbon trajectories for a representative 500 m^2 Lagos residential building over a 20-year period across three construction quality scenarios parameterized by QSII scores. Monte Carlo simulation (5,000 iterations) propagated uncertainty from maintenance record gaps through carbon projections, yielding 95% confidence intervals of ±14% across all scenarios (Table 4, Fig. 3). Results indicate a 44% lifecycle carbon difference between high-quality (QSII > 80) and poor-quality (QSII < 50) construction scenarios substantially exceeding the 3–4% initial cost premium associated with quality-first practices. Sensitivity analysis of emission differentials attributed 42% of this variance to repair frequency, 38% to material selection, and 20% to conceptual operational energy penalties from thermal degradation.

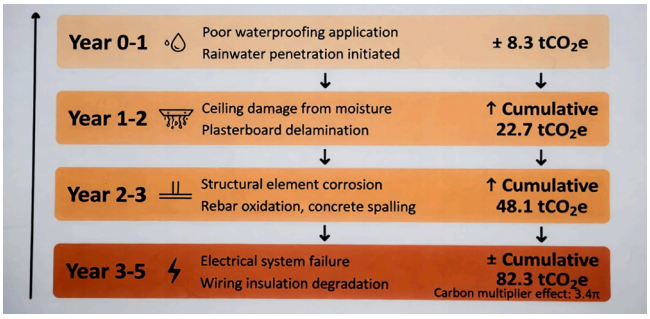


Fig. 2. Waterproofing defect propagation pathway showing cascading secondary damage chains and cumulative embodied carbon accumulation over the five-year observation window (tCO_2e per 1,000 m^2).

Table 4
Conceptual digital twin prototype — lifecycle carbon projections by quality scenario (500 m^2 residential building, 20-year period, Monte Carlo 5,000 iterations).

Quality Scenario	QSII range	20-yr carbon (tCO_2e)	95% CI (±14%)	vs. High-quality baseline
High quality (quality-first)	QSII > 80	247	±34.6	Baseline
Conventional construction	QSII 60–70	302	±42.3	+22% (+55 tCO_2e)
Poor quality construction	QSII < 50	356	±49.8	+44% (+109 tCO_2e)

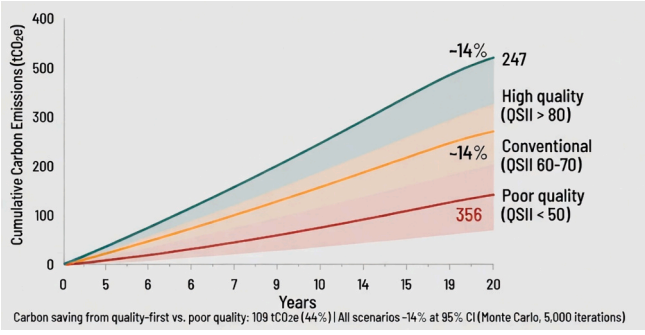


Fig. 3. Lifecycle carbon emission trajectories for three QSII-based quality scenarios over 20 years (500 m^2 residential building, Lagos). Error bands represent 95% confidence intervals from Monte Carlo simulation (±14%).

These findings challenge the conventional assumption that material selection is the primary driver of lifecycle carbon, demonstrating instead that defect prevention and maintenance frequency are equally significant sustainability determinants. The conceptual Digital Twin prototype thus provides a structured environment for visualizing how early quality investments yield disproportionate long-term carbon reductions, while explicitly acknowledging its current role as a descriptive scenario tool

Table 3
Defect propagation pathways and cumulative embodied carbon impacts for the two highest-priority QSII defect categories across a five-year observation window (tCO_2e per 1,000 m2 floor area, ICE v3.0 reference factors)..

Time period	Primary defect	Secondary effect	Repair required	Cum. carbon (tCO_2e/1000 m^2)
Yr 0–1	Poor waterproofing application	Rainwater penetration initiated	Sealant reapplication	8.3
Yr 1–2	Ceiling damage from moisture	Plasterboard delamination	Ceiling board replacement	22.7
Yr 2–3	Structural element corrosion	Rebar oxidation, concrete spalling	Patch concrete repairs	48.1
Yr 3–5	Electrical system failure	Wiring insulation degradation	Rewiring sections	82.3
Yr 0–3 (cover)	Insufficient rebar cover	No visible symptom (latent)	None initially	0
Yr 3–5 (cover)	Surface cracking & rusting	Section loss, structural risk	Structural repair	67.8

rather than an operational predictive system (Table 4, Fig. 3).

Fig. 3 illustrates these three trajectories, demonstrating the progressive divergence of cumulative carbon emissions between quality scenarios over the 20-year period.

The sensitivity analysis indicated that, of the carbon difference erected between the building and the reference, 42% was due to the frequency of repair; 38% was due of materials selected; and 20% was due to the operational energy penalties or thermal defects, therefore the existing convention that material selection is the largest contributor to the environmental impact of new buildings was incorrect, with defect prevention and maintenance being similarly significant carbon determinants.

5. Discussion

5.1. Theoretical contributions

The research contributes to advancing construction sustainability through three different theoretical contributions that are each addressed within the framework of gaps found in the literature that pertain to the three contributions in Section 2.

The first contribution is an advancement of the theory of construction defect severity via the introduction of a new conceptual type of defect known as a 'carbon-amplifying defect'; by definition, these types of defects do not fit perfectly within the defined structural severity criteria. As referenced in Section 2.1, the gap within previous literature was that there was no framework linking defects, severity, and the long-term carbon and energy implications of these types of defects in developing economy countries that are classified as hot-humid climates. The gap in current literature can now be closed by the empirical research provided by this study, demonstrating that the carbon multiplier effects associated with the carbon-amplifying defect (ranging from 1.6× to 3.4×) systematically reorder the defect priority hierarchy based on the inclusion of sustainability-type criteria, meaning the existing conventional defect severity ranking developed by Okuntade [4] materially under-represents the sustainability liabilities associated with carbon-amplifying defect types. This is not merely an improvement to an existing theory; it is a complete reconceptualization of what it means for severity to be evaluated against long-term performance of a building rather than immediate structural risk.

Secondly, this research contributes to the advancement of the quality-sustainability integration (QSII) instrument by introducing it as an empirically calibrated, operationally viable data-limited framework for use in developing countries as called for by Zhang et al. [10], where they identify a gap in the current research literature surrounding integrated frameworks applicable to those types of economies. Current sustainability indices, including the aggregated-composite methods discussed by Yang et al. [17], are mostly developed with a consideration of having access to high-quality simulation inputs and/or longitudinally monitored data. In contrast, the QSII was designed and built with the realities of operating in an environment with extremely limited data availability due to both its use of expert-concurrent calibration (rather than instrumented monitoring) and its use of documented uncertainty as an uncertainty input to the model (rather than as a limitation that needs to be concealed). This methodological reframing creates a new theoretical method for assessing the sustainable development amounts of emerging economies on a regional basis using the QSII as a common, standards-based facility for assessing sustainable development in developing countries.

Theoretical relationships between lifecycle performance of buildings and construction management quality control and construction management are strengthened because building repair is the major factor causing carbon differences between scenarios (42% to 38%) rather than material selection. This clearly fills the gap identified in Section 2.2 of the LCA literature regarding failure-driven turnover of construction materials not being properly accounted for in lifecycle carbon models. This research creates a framework for conducting future studies of the relationship between local knowledge and developing theories to better understand collaborative sustainable development practices based on Transaction Cost Economics and Circular Economic principles.

5.2. Comparison with related research

The carbon multiplier ranges (1.6x to 3.4x) identified in this study support the common principle that the total amount of embodied carbon from repair cycles is significantly greater than just the direct material carbon emitted in defect-prone building systems [20]. However, very few studies have used carbon multiplier methodologies in warm/humid countries, which limits the number of direct comparisons to be made to other countries/regions (this is a critical contribution of this study because it identifies a knowledge gap in the methodology used).

The excess lifecycle carbon associated with waterproofing defects is much higher (28%) than what would typically be experienced due to performance penalties associated with similar types of envelope failures in temperate climates because moisture-driven deterioration mechanisms tend to occur at a significantly faster rate in humid climates like Lagos (consistent with [16]). This also emphasizes the importance of developing defect management systems that are specific to climate zones when developing building codes and considering how to weigh the construction quality vs. materials used within a building code. The amount of carbon attributed to the frequency of repairs (42%) creates substantial opportunity for reformulating how the LCA process is applied, and creates opportunities for the building industry to implement innovative defect management practices.

5.3. Practical implications

5.3.1. For construction professionals

The QSII provides a usable spreadsheet tool to evaluate the sustainability effects of building defects through data present to a construction industry professional. The defect propagation pathways (Table 3 and Fig. 2) provide an abstract model of lifecycle concepts, showing, in a graphical format, how latent defects will show up in the form of sustainability liabilities. The case study analysis showed that remediation of a waterproofing deficiency after the fact took 3–5 times as many resources as could have been spent on preventing the deficiency at the time of construction. The quality-first building example produced 18–29% less carbon emissions over its lifecycle and 22–31% less cost to maintain over a 15-year period, with carbon reduction paybacks of 12–18 months.

5.3.2. For policy and regulatory development

Incorporating quality-adjusted carbon factors into current life cycle assessment databases would allow government agencies to generate more accurate environmental product declarations based on actual performance during use, rather than idealised laboratory performance. Integrating quality service indicator (QSI) metrics into assessment systems for buildings and public procurement may facilitate the institutionalisation of defect prevention as part of a sustainable strategy and treat quality assurance and sustainability as one regulatory system, rather than two discrete, separate systems.

5.3.3. For digital implementation strategy

The Digital Twin model illustrates the correct order for adopting technological innovations in environments where there are limited resources. It starts with providing stakeholders with ways to visualize and communicate scenarios; by first improving their awareness of the system, they will be ready for operational deployment of predictive systems. Additionally, the use of productized, standardized (industry-standard) software such as BIM-LCA (Building Information Modeling Life-Cycle Assessment) tools like Revit and Tally rather than bespoke software, helps enhance the methodological robustness of the

deliverables being produced while providing the ability to apply practical (i.e., usable) tools. The focus on quantifying uncertainty (i.e., reporting 95% confidence intervals [±14%] and publishing parameters used to determine each probability) will create a template for honest digital construction research and inform the general public about limitations in the accuracy of the data being reported and the level of accuracy of the analyses that have been performed.

5.4. Limitations and boundary conditions

Geographic scope represents a key boundary condition: data collection occurred exclusively in Lagos, Nigeria. While this provides strong internal validity for the local context, transferability to arid, temperate, or other tropical sub-climates cannot be assumed without empirical verification. The hot-humid climate accelerates moisture-driven degradation mechanisms; carbon multiplier values for envelope defects, for instance, may differ substantially in regions where precipitation patterns and cooling demands vary. Therefore, one cannot assume that a direct transfer can be made to an arid or temperate or other tropical sub-climates, without specifically calibrating for their respective climates. The maintenance dataset consists of only five years of observations so the information will not fully capture the long-term propagation pathways of slowly developing defects and therefore cannot be used as a model for the carbon contribution from the lifecycle of a defect at the end of its lifecycle. Since ICE v3.0 emission factors were used as proxies for carbon intensities of the supply chain in Nigeria, there is an assumption that the absolute value of emissions may vary from the accurate representation; however, the sensitivity analysis to rank defects was shown to be robust for relative defector rankings. The QSII weighting was confirmed through structured consensus workshops, and while it does contain subjectivity, there is no empirical validation for the Conceptual digital Twin prototype or operationally validating as it is presently a conceptual tool only.

6. Conclusion

In this study, we provide empirical support that construction quality management is a latent but quantifiable lever for improving energy and carbon performance in buildings. Using data collected from Lagos, Nigeria, we illustrate how traditional ways of estimating the severity of defects have not taken into account the sustainability liabilities associated with defects that increase carbon emissions. By developing the Quality/Sustainability Impact Index (QSII), we demonstrate a significantly different order of prioritizing defects than through structural severity alone; envelope and waterproofing defects are now ranked as the top sustainability risks because of their high carbon multiplier (up to 3.4×) and high frequency of repair cycle. Equally, longitudinal case studies show that the triggering of waterproofing failures causes secondary damage to the building and accumulates 82.3 tCO_2e of emissions per 1,000 m^2 over a five-year period, while envelope defect cases generate 28% more lifecycle carbon than comparable cases with quality control measures in place. The scenario analysis shows that the lifecycle carbon difference between high-quality and low-quality construction scenarios is 44%, well in excess of the 3–4% increase in initial construction costs associated with quality-first practices. Most importantly, 42% of the lifecycle carbon difference can be attributed to frequency of repair rather than simply selection of materials, disputing common assumptions about lifecycle carbon.

The geographical and climatic conditions define the parameters of this research. Therefore, all findings, carbon multipliers and defect propagation pathways have been developed with respect to Lagos' hot humid climate, local material supply chains and construction practices used in institutional settings. The moisture-induced accelerated degradation mechanisms described in Lagos may not apply in the same way to arid climates or to temperate or other tropical sub-climates. Thus, while the QSII methodology can be used in different locations; the quantitative outputs of the QSI method must first be validated using local data, and recalibrated with local parameters prior to using the QSI method in other locations.

Future research should place priority on the development of Environmental Product Declarations in Nigeria (to replace the Proxy Emission Factor), long-term monitoring of 15–20 years to reflect the entirety of defect propagation lifecycles; geographical verification studies to adapt the QSII parameters to the range of climate zone; and the progressive operationalization of the Digital Twin conceptual prototype as the building performance monitoring infrastructure matures within emerging economies. By reconceptualising defects in construction as liabilities to systemic sustainability, this study offers a practical and context-appropriate framework for integrating carbon reduction and quality assurance within resource-constrained environments.

CRediT authorship contribution statement

Tope Femi Okuntade: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Olatunji Abisuga:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Data collection instruments

Survey Instrument Excerpt: For each defect category, rate the following:

(1) Severity impact on building maintenance (1–4 scale matching Okuntade)
(2) Carbon emission impact (1–5 scale: 1 = negligible, 5 = severe)
(3) Estimated annual repair frequency (0.1–5.0 cycles/year)

Case study data collection protocol

- Initial defect documentation from snagging reports
- Annual maintenance records (materials, labor, costs)
- Post-repair inspection reports
- Material quantity documentation for repairs
- Photographic evidence of defect progression

Appendix B. Ethical considerations

This research adhered to strict ethical protocols:

- Written consent obtained from all survey participants
- Building owners granted explicit permission for case study inclusion
- All identifying information anonymized in reporting
- Raw data stored on password-protected university servers
- Data retention period limited to 5 years post-publication

Appendix C. Quality-sustainability integration index (QSII) calculation example

For "Inadequate waterproofing and drainage":

- Traditional Severity Index (SI): 78.3 (from survey)
- Carbon Emission Index (CEI): 84.7 (from expert validation)
- Repair Frequency Index (RFI): 3.2 repairs/year (from case studies)
- RFI was normalised to a 0–100 scale using min–max normalisation across observed defect frequencies, ensuring dimensional consistency with SI and CEI
- RFI (normalised 0–100 scale): 64.0
- QSII = (78.3 × 0.5) + (84.7 × 0.3) + (64.0 × 0.2) = 86.2

Data availability

The Authors state that any data that is collected through the Authors' study (as outlined above) can be made available upon request from the primary author for reasonable requests. All data is aggregated and does not identify individual subjects.

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